Kihwan Kim

juljin1875@gmail.com
kihwan.kim · github.com/simulacre7
Seoul, South Korea

Software Engineer · 7 years — deploying complex AI systems for the people who use them

Engineer with 7 years of experience taking complex AI and analytics systems and deploying them in a form the people who actually use them can trust. My background is in HCI and Visual Analytics research on human–AI collaboration; since then I have embedded with operations teams and non-expert users to turn scattered tools and model outputs into end-to-end solutions — from an AutoML platform for non-experts to an agent-based operations automation built on a browser-driving agent. I own problems from discovery through production, and extend the tools I build on when they fall short.

Highlights

- **End-to-end AI deployment** — embedded with an operations team, scoped their cross-system workflow, and deployed an agent-based automation that cut hours of investigation to minutes (Levvels, PageAgent)
- **Making AI usable for non-experts** — built codeless studio flows and explainable-AI visualizations so people without ML background could operate an AutoML platform (TmaxEnterprise)
- **Extending tools for production** — extended an open-source browser agent with a selector strategy and a human-in-the-loop escape hatch so automations fail loudly to a person, not silently
- **Owning across the stack** — contributed 10 merged PRs to an open-source MMORPG, from the TypeScript client into the Rust server, using a builder/reviewer coding-agent workflow with output verification

Experience

Levvels — Frontend Engineer
Jul 2025 – Present

AI-native product development and internal operations automation. I use executable prototypes and browser automation to reduce the cost of product changes and repeated investigation work.

PageAgent-based Operations Automation `AI deployment · end-to-end`

- Embedded with the operations team to understand how they carried state across several systems (customer chat, delivery tracking, internal admin, external email) to make a single decision
- Designed and deployed an agent-based automation that turned hours of manual investigation into minutes, connecting the needed context into one verifiable result view
- Extended Alibaba's open-source PageAgent for a Chrome Managed Profile: injected sitemaps and DOM selectors so the agent relied less on per-screen visual reasoning and moved faster
- Built a human-in-the-loop escape hatch (Slack hand-off) the base tool lacked, so a stuck automation fails loudly to a person instead of returning a wrong result silently
- Separated LLM output from reproducible action sequences and result presentation, so operators can re-verify each conclusion against its source

AI Character Chat `streaming`

- Separated in-progress streamed responses from persisted history; designed the send / complete / fail / regenerate state machine for interrupt-and-retry-heavy LLM responses
- Built authoring and review tools for stories, lorebooks, and search keywords

React · Next.js · TypeScript · TanStack Query · Zustand · SSE · Playwright · Sentry · PageAgent · Generative UI

RIDI Corp. — Frontend Engineer
May 2022 – Jul 2025

Decoupled product experimentation and operational UI changes from deployment cycles on a large content platform.

Server-driven UI & large-scale rendering

- Rendered server-declared UI on the client to decouple screen changes from deploys; migrated PHP pages toward React incrementally to lower risk
- Optimized large-list rendering with windowing, dynamic-height measurement, and scroll restoration; analyzed and reduced frame drops from large DOM trees
- Tracked real-user errors with Sentry and stabilized mobile, desktop, and special-device environments

React · Next.js · TypeScript · Server-driven UI · Sentry · Virtualization

TmaxEnterprise — Research Engineer
Feb 2020 – May 2022

Designed flows, visualizations, and interactions that let non-expert users work with complex AI and analytics — usability and explainability first, not technology first.

AutoML Platform explainable AI

- Designed and implemented a codeless studio so users without model-development experience could build and operate models
- Developed explainable-AI visualizations and dashboards that show the reasoning beside the prediction, so a non-expert has a basis to act
- Researched the interface between AutoML engines and model developers; clarified requirements for an early-stage platform

React · TypeScript · Material-UI · Python

Open Source

OpenMMO — Contributor
Jul 2026 – Present

Open-source MMORPG where AI agents and humans play through the same protocol (Rust · Svelte · WebAssembly).

- 10 PRs merged, owning changes end-to-end from the TypeScript client into the Rust server — cache eviction, account bans, and trade validation
- Paired with a coding agent under a builder/reviewer workflow: the builder writes, a separate model reviews, and behavior is pinned by tests before submission
- Wrote the project's CONTRIBUTING guide (CI checks and how to pick work), linked from all three README translations

vite-plus — Contributor
Aug 2026

Toolchain from VoidZero, the team behind Vite and Vitest (Rust · TypeScript).

- Fixed a shell-integration gap in Rust where the global working-directory flag (-C) was dropped before the env command, so the change never reached the current shell; kept it consistent across zsh, Fish, Nushell, and PowerShell and pinned the templates with snapshot tests (#2508)
- In the vite-task task runner, bulk env queries were validated against the unfiltered environment, so per-run CI variables broke caching on every GitHub Actions run; extended the untrackedEnv contract to bulk queries with unit and e2e coverage (#693, in review)

Education

Ulsan National Institute of Science and Technology (UNIST)
2013 – 2020

M.S. in Computer Science (HCI · Visual Analytics; human–AI collaboration).
B.S. in Technology Management (CS multi-disciplinary major).

- “An Empirical Analysis on Transparent Algorithmic Exploration in Recommender Systems” — **CoRR 2021** · arxiv.org/abs/2108.00151: measured how disclosing algorithmic exploration affects user trust and feedback quality
- “Modeling Exploration/Exploitation Decisions through Mobile Sensing for Understanding Mechanisms of Addiction” — **MobiSys '19** · dl.acm.org/doi/10.1145/3307334.3328599: estimated user reward functions from mobile-sensing behavior logs via inverse reinforcement learning

Full CV, case studies, and portfolio at kihwan.kim.